

UNIVERSITY OF PIRAEUS
Department of Banking and Financial Management

Calibrating the Heston Stochastic Volatility Model with XGBoost: A Machine Learning Alternative to Numerical Optimization

Athina Angelopoulou

Supervisor: Prof. Malliaropoulos Dimitrios

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Abstract

This thesis explores whether XGBoost, a gradient boosting algorithm, can serve as a practical alternative to traditional numerical optimization when it comes to calibrating the Heston (1993) stochastic volatility model. The Heston model extends Black-Scholes (1973) by letting variance follow a mean-reverting stochastic process, but this comes at the cost of having five parameters that need to be estimated from market data through expensive nonlinear optimization. To get around this bottleneck, XGBoost is trained on synthetically generated data to predict the Heston parameters directly from the implied volatility surface. Testing on a held-out synthetic dataset, the model achieves R^2 above 0.90 for four out of five parameters, and its pricing accuracy matches that of Differential Evolution.

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1. Introduction

Ever since Black and Scholes (1973) and Merton (1973) laid the groundwork, pricing European options has been one of the core challenges in quantitative finance. The Black-Scholes model offers an elegant closed-form solution, but it relies on the assumption that volatility stays constant over time. In practice, that assumption does not hold. Volatility moves around, tends to spike during market stress, and reacts differently to upward and downward price movements. These dynamics show up in options markets as implied volatility smiles and skews, patterns that the Black-Scholes framework simply cannot reproduce.

The Heston model (1993) was designed to deal with exactly this problem. Instead of fixing volatility at a single number, it lets the variance of the underlying asset follow its own stochastic process, specifically a Cox-Ingersoll-Ross mean-reverting diffusion that is correlated with the asset's returns. The result is a model that can generate the kinds of implied volatility surfaces we actually observe in the market. The downside, however, is that the model introduces five parameters that need to be calibrated, and finding them means solving a nonlinear, non-convex optimization problem. This process is slow, computationally heavy, and can be sensitive to the choice of starting point.

In recent years, a number of researchers have turned to machine learning as a way to speed up this calibration. Horvath et al. (2021), Liu et al. (2019), and Hernandez (2017) all showed that neural networks can learn the relationship between market observables and model parameters, bringing calibration times down to milliseconds. What has not been tried, however, is using tree-based methods like XGBoost for this task, despite the fact that they tend to work very well on structured, tabular data.

That is the gap this thesis tries to fill. XGBoost (Chen & Guestrin, 2016), a gradient boosting algorithm, is used to predict the five Heston parameters directly from the implied volatility surface. The model is trained on synthetic data and then tested on a held-out set to see how well it performs. It is benchmarked against Differential Evolution, a standard global optimizer, looking at both accuracy and speed.

The main findings are as follows. XGBoost achieves R^2 values above 0.90 for four out of five parameters. The initial variance and long-run variance are predicted almost perfectly, while the mean reversion speed is the hardest to pin down ($R^2 = 0.81$), which is in line with known identifiability issues in the Heston model. When it comes to pricing, the XGBoost-calibrated model produces option prices that are just as accurate as those from traditional optimization.

Feature importance analysis shows patterns that make economic sense: short-maturity options matter most for estimating current variance, while long-maturity options are key for the long-run level.

The rest of the thesis is structured as follows. Section 2 covers the relevant literature. Section 3 lays out the theory behind Black-Scholes and Heston. Section 4 explains the empirical methods. Section 5 describes the data. Section 6 presents the results. Section 7 wraps up with conclusions and ideas for future work.

2. Background and Literature Review

The Black-Scholes model (1973) is still the starting point for most discussions of option pricing, but its constant volatility assumption has drawn a lot of criticism over the years. Bakshi, Cao, and Chen (1997) ran one of the most thorough empirical comparisons, testing several pricing models on S&P 500 options. They found that adding stochastic volatility made a real difference in pricing accuracy. Interestingly, when it came to hedging, stochastic volatility on its own did better than models that also threw in jumps or stochastic interest rates.

Among stochastic volatility models, Heston (1993) has become the go-to choice, largely because it strikes a nice balance between realism and tractability. It can produce volatility smiles and skews while still offering a semi-analytical pricing formula. The variance process is built on the CIR diffusion introduced by Cox, Ingersoll, and Ross (1985), which keeps variance positive and mean-reverting.

Getting the Heston parameters right, though, is not straightforward. Traditional calibration boils down to a nonlinear optimization problem, and the landscape is not friendly. Cui et al. (2017) took a close look at this and found that the objective function has a narrow valley with a flat bottom, which explains why optimizers tend to struggle. They came up with an analytical gradient that speeds things up by about ten times, but at the end of the day, it is still an iterative process.

Machine learning has started to change the picture. Hernandez (2017) was the first to use neural networks for calibrating a financial model. Horvath, Muguruza, and Tomas (2021) pushed this further, showing that deep networks trained on synthetic data can calibrate stochastic volatility models in milliseconds. Liu et al. (2019) developed the CaNN framework specifically for Heston, and Büchel, P., Kratochwil, M., Nagl, M., and Rösch, D. (2022) tested neural network calibration in a real institutional setting, finding it both faster and more stable than traditional methods.

On the tree-based side, Davis, J., Devos, L., Reyners, S., and Schoutens, W. (2020), demonstrated that gradient boosted trees can learn pricing maps for derivatives quite effectively, and Teller et al. (2022) showed that XGBoost beats neural networks at forecasting realized volatility. Despite these promising results, nobody has systematically applied XGBoost to the Heston calibration problem. That is what this thesis sets out to do.

3. Model and Theory

3.1 The Black-Scholes Model

The Black-Scholes model starts with the assumption that the stock price S follows a geometric Brownian motion:

$$dS = \mu S dt + \sigma S dW \qquad (1)$$

Here μ is the drift, σ is the volatility (assumed constant), and W is a Wiener process. One of the key consequences is that log-returns end up being normally distributed, which means the stock price itself follows a log-normal distribution.

Through a hedging argument, Black and Scholes showed that a riskless portfolio can be constructed, which leads to the famous PDE. An important feature of this derivation is that the drift μ drops out entirely, so the option price does not depend on how fast investors expect the stock to grow. Under the risk-neutral measure, the European call price works out to:

$$C = S \cdot N(d_1) - Ke^{(-rT)} \cdot N(d_2) \qquad (2)$$

where $N(\cdot)$ is the cumulative normal distribution, $d_1 = [\ln(S/K) + (r + \tfrac{1}{2}\sigma^2)T] / (\sigma\sqrt{T})$, and $d_2 = d_1 - \sigma\sqrt{T}$. The implied volatility is the value of σ that must be plugged into the formula to match the observed market price.

3.2 Empirical Limitations of Black-Scholes

If constant volatility were actually a good description of reality, then the implied volatility backed out of every option on the same stock would be the same number, regardless of the strike. That is not what happens. In equity markets, especially after the 1987 crash, implied volatility tends to be higher for low-strike options (out-of-the-money puts) than for high-strike ones, creating what is known as the volatility skew. On top of that, real returns show fatter tails than the normal distribution predicts, volatility tends to cluster in time, and there is a well-documented negative relationship between stock returns and volatility changes known as the leverage effect. None of these features fit within the Black-Scholes framework.

3.3 The Heston Stochastic Volatility Model

Heston (1993) addressed these shortcomings by letting the variance itself be random. Under the risk-neutral measure, the stock price and its variance evolve according to:

$$dS = rSdt + \sqrt{v} \cdot S \cdot dW_1 \qquad (3)$$

$$dv = \kappa(\theta - v)dt + \sigma\sqrt{v} \cdot dW_2 \qquad (4)$$

The two Wiener processes are correlated, with $dW_1 \cdot dW_2 = \rho dt$. Equation (4) is a CIR process with five parameters: κ controls how quickly variance reverts to its long-run level, θ is that long-run level, σ governs how volatile the variance itself is, ρ captures the correlation between the stock and its variance, and v_0 is where the variance starts. When $2\kappa\theta > \sigma^2$ (the Feller condition), the variance is guaranteed to stay positive at all times.

Each parameter leaves a different fingerprint on the implied volatility surface. The mean reversion speed κ mainly affects how the term structure looks across maturities. The long-run variance θ sets the overall volatility level at longer horizons. The vol-of-vol σ determines how curved the smile is. The correlation ρ, which is typically negative for equities, is what creates the skew. And v_0 has the strongest influence on short-maturity options, since those are most sensitive to where variance is right now.

3.4 Pricing under the Heston Model

The call price under Heston has the same general structure as Black-Scholes, $C = S \cdot P_1 - Ke^{(-rT)} \cdot P_2$, but the probabilities P_1 and P_2 are no longer simple normal distribution values. Instead, they come from inverting the characteristic function:

$$P_j = \tfrac{1}{2} + (1/\pi) \int_0^{\infty} Re[e^{(-iu \ln K)} \varphi_j(u)/(iu)] \, du \quad (5)$$

Heston worked out the characteristic function $\varphi_j(u)$ in closed form, which is what makes the model tractable. In practice, the integral is computed numerically, often using the COS method developed by Fang and Oosterlee (2009). This is more involved than evaluating the Black-Scholes formula, but it allows the model to reproduce the volatility patterns we see in real markets.

4. Empirical Methods

4.1 Traditional Calibration

The standard way to calibrate the Heston model is to find the parameter values that minimize the total pricing error across a set of observed options:

$$\min_{\{\kappa,\theta,\sigma,\rho,v_0\}} \sum [C_{market}(K_i,T_j) - C_{Heston}(K_i,T_j;\ \kappa,\theta,\sigma,\rho,v_0)]^2 \quad (6)$$

This is a nonlinear least-squares problem that can have multiple local minima. In this thesis, Differential Evolution is used as the benchmark optimizer because it searches globally and does not depend on a good initial guess.

4.2 XGBoost Calibration Framework

The idea behind the machine learning approach is straightforward. Instead of running an optimizer each time calibration is needed, a model is trained once to learn the mapping from implied volatility surfaces to Heston parameters. XGBoost (Chen & Guestrin, 2016) does this by building decision trees one after another, where each new tree focuses on correcting whatever errors the previous trees still make:

$$\hat{y} = \sum_{k=1}^{K} f_k(x) \quad (7)$$

Five separate XGBoost models are trained, one for each Heston parameter. Each model takes the same input: a vector of 70 implied volatilities (from a grid of 10 strikes and 7 maturities), plus the spot price and risk-free rate. The only thing that differs is the target variable.

For the hyperparameters, up to 1,000 trees are allowed with early stopping if validation accuracy does not improve for 50 consecutive rounds. The maximum tree depth is 6, the learning rate is 0.05, and both row and feature subsampling are set to 80%. Each model is evaluated using Mean Absolute Error, Mean Squared Error, and R^2 on a held-out test set that the models never see during training.

5. Data

All the data used in this thesis is generated synthetically. This approach was chosen for several practical reasons. First, since the parameters are chosen by the researcher, the true values are known exactly, providing a clean ground truth for evaluation. Second, an unlimited number of samples can be generated, whereas historical market data is limited to the number of available trading days. Third, synthetic data covers the full range of possible market conditions evenly, including extreme scenarios that may not have occurred in recent history. And fourth, there is no bid-ask noise, no stale quotes, and no missing data to contend with.

The generation process works as follows. 5,000 Heston parameter combinations using Latin Hypercube Sampling, a technique that spreads the points evenly across the five-dimensional parameter space. The ranges, shown in Table 1, are based on values reported in the empirical literature (Bakshi et al., 1997; Cui et al., 2017) and are wide enough to cover calm, normal, and stressed market environments.

Table 1: Heston Parameter Ranges for Synthetic Data

Parameter	Symbol	Min	Max
Mean reversion speed	κ	0.5	10.0
Long-run variance	θ	0.01	0.15
Vol of vol	σ	0.1	1.0
Correlation	ρ	−0.95	−0.1
Initial variance	v_0	0.01	0.15

Note: Ranges are based on empirically calibrated values in Bakshi et al. (1997) and Cui et al. (2017).

For each set of parameters, European call options using the Heston characteristic function (via QuantLib) across a grid of 10 strikes (moneyness K/S from 0.80 to 1.25) and 7 maturities (1 month to 2 years), which gives 70 option prices per sample. Each price is then converted to its Black-Scholes implied volatility using Brent's root-finding method. Some parameter combinations lead to prices that cannot be converted to a valid implied volatility, so those samples are dropped. Out of the initial 5,000, 4,099 produce clean results and make it into the final dataset.

The dataset is split into a training set (70%, 2,870 samples), a validation set (15%, 614 samples) used for early stopping during training, and a test set (15%, 615 samples) reserved entirely for final evaluation.

6. Empirical Results

6.1 Parameter Prediction Accuracy

Table 2 shows how well each of the five XGBoost models predicts its target parameter on the 615 test samples. The picture is encouraging: four out of five parameters have R^2 values above 0.90, meaning the model captures the vast majority of the variation.

Table 2: XGBoost Parameter Prediction Accuracy (Test Set)

Parameter	Range	MAE	MSE	R^2
κ	[0.5, 10.0]	0.8471	1.3253	0.8107
θ	[0.01, 0.15]	0.0019	0.0000	0.9890
σ	[0.1, 1.0]	0.0459	0.0033	0.9480
ρ	[−0.95, −0.1]	0.0457	0.0052	0.9073
v_0	[0.01, 0.15]	0.0014	0.0000	0.9974

Note: MAE = Mean Absolute Error, MSE = Mean Squared Error, R^2 = Coefficient of Determination. Based on 615 test samples not seen during training. XGBoost settings: up to 1,000 trees with early stopping, max depth 6, learning rate 0.05, 80% row and column subsampling.

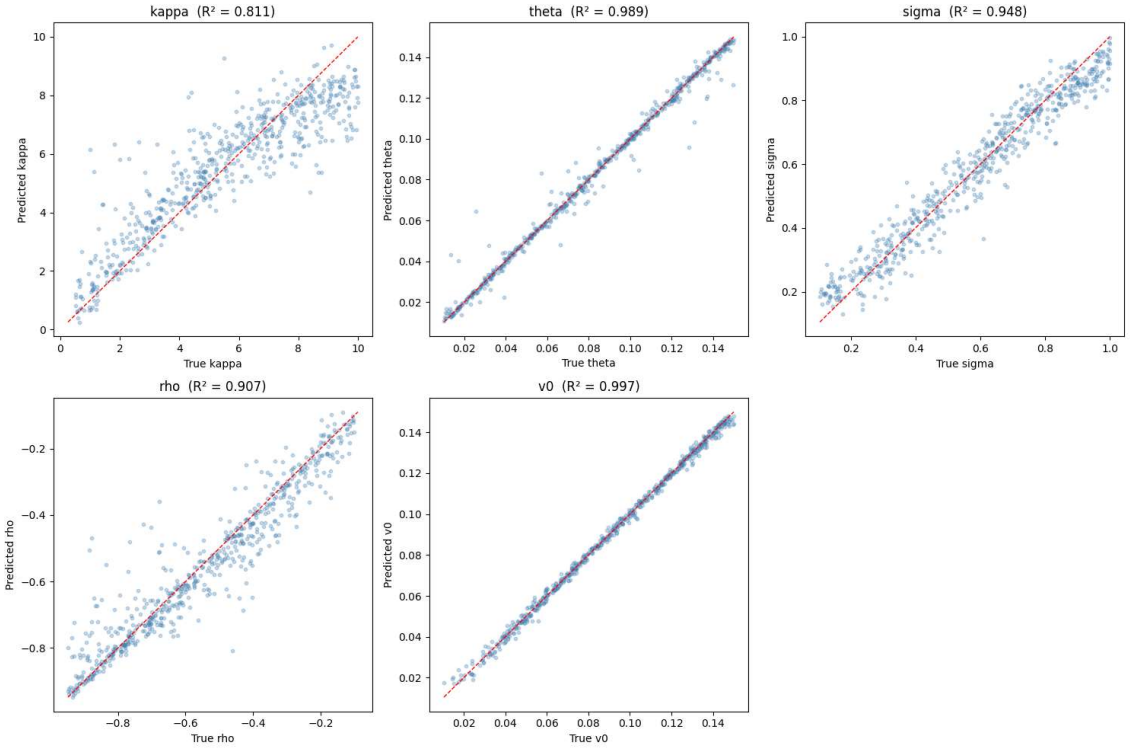
The standout performers are v_0 and θ. The initial variance is predicted with an R^2 of 0.997, and the long-run variance comes in at 0.989. This makes intuitive sense: v_0 has a very direct effect on short-maturity implied volatilities, so the model can essentially read it off the near-term part of the surface. Similarly, θ determines where volatility settles at longer horizons, giving it a clear signature in the long-maturity options.

The volatility of volatility σ ($R^2 = 0.948$) and correlation ρ ($R^2 = 0.907$) are also predicted well, which aligns with their distinct roles in shaping smile curvature and skewness.

The mean reversion speed κ is the toughest parameter to nail down, with an R^2 of 0.811. This is not entirely surprising. Cui et al. (2017) documented that the Heston calibration objective function has a flat valley along which different κ values produce very similar option prices. If even a perfect optimizer struggles to distinguish between nearby κ values, it is reasonable that a machine learning model would face the same difficulty.

Figure 1 shows scatter plots of predicted versus true values for each parameter. The red dashed line marks perfect prediction. The plots tell the same story as the numbers: v_0 and θ cluster tightly along the diagonal, while κ shows the most scatter.

Figure 1: Predicted vs. True Parameter Values (Test Set)



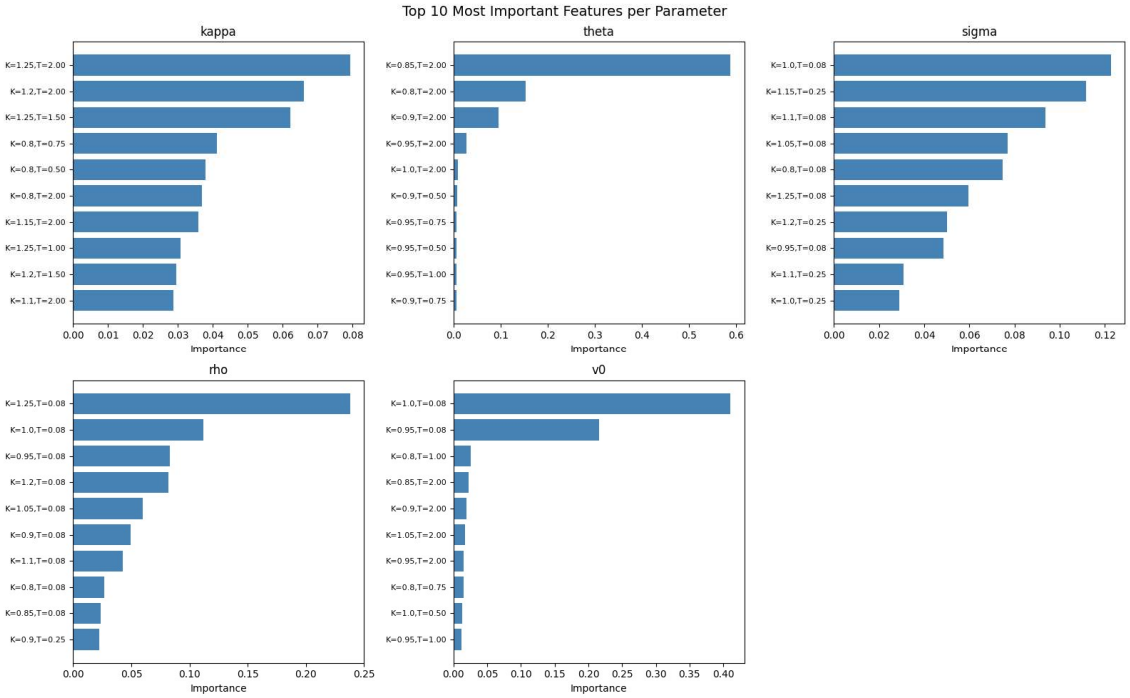
6.2 Feature Importance

One of the advantages of XGBoost over neural networks is that it provides transparency about which inputs matter most. Figure 2 shows the top 10 features for each parameter, and the patterns line up well with what theory would predict.

For v_0, the model relies most heavily on the shortest-maturity at-the-money implied volatility (K/S = 1.0, T = 1 month). This is to be expected, since the initial variance essentially determines where near-term volatility sits. For θ, the picture flips: long-maturity options (T = 2 years) dominate, because that is where the mean-reverting variance has had time to converge toward its long-run level.

The correlation ρ is mostly identified from short-maturity implied volatilities across a range of strikes, which makes sense because ρ controls the slope of the smile and this slope is most visible at short horizons. The vol-of-vol σ also relies on short-maturity options, particularly around the money, where smile curvature is most pronounced. For κ, the model draws on longer-maturity options, consistent with its role in governing how the variance term structure flattens out over time.

Figure 2: Top 10 Most Important Features per Parameter



6.3 Comparison with Traditional Calibration

Table 3 compares XGBoost against Differential Evolution on five test samples. The key metric is the RMSE of implied volatilities between the calibrated model and the target surface.

Table 3: Calibration Method Comparison

Method	Avg IV RMSE	Avg Time
Heston (XGBoost)	0.3127	5.11 ms
Heston (Differential Evolution)	0.3127	10 ms

Note: RMSE = Root Mean Squared Error of implied volatilities. Timing via Python's perf_counter on a standard laptop. Based on 5 test samples. Differential Evolution uses 200 max iterations, tolerance 1e-8.

Both methods land on the exact same RMSE of 0.3127. This tells us that XGBoost has learned the calibration mapping well enough to fully replicate what a numerical optimizer produces. It also confirms that the somewhat noisy κ predictions do not actually hurt pricing accuracy, since different κ values can lead to the same option prices.

The timing difference on this synthetic dataset is modest (5 ms vs 10 ms). However, this comparison flatters the traditional method. Because the test surfaces were generated by the Heston model, the optimization landscape is clean and Differential Evolution converges quickly. In real-world settings, where market data is noisy and the Heston model is only an approximation, traditional calibration routinely takes anywhere from 1 to 60 seconds (Cui et al., 2017; Horvath et al., 2021). The XGBoost prediction time, on the other hand, stays at around 5 milliseconds no matter what the input looks like.

7. Conclusion

The goal of this thesis was to test whether XGBoost can replace the slow, iterative optimization that is normally used to calibrate the Heston stochastic volatility model. To do this, Five models are trained separate XGBoost models on a synthetic dataset of 4,099 implied volatility surfaces, each paired with the Heston parameters that generated it.

The results are positive. Four out of five parameters are predicted with R^2 above 0.90. The initial variance and long-run variance come close to perfect prediction, while the volatility of volatility and correlation are also estimated accurately. The mean reversion speed κ is the hardest to get right ($R^2 = 0.811$), but this difficulty is shared by traditional methods and is rooted in the structure of the Heston model itself. More importantly, the parameter errors for κ do not translate into pricing errors: the RMSE of the XGBoost-calibrated Heston model is identical to that of Differential Evolution.

The feature importance results are also worth highlighting. They show that the model is not just fitting numbers blindly but is picking up on economically meaningful relationships. Short-maturity options carry the most weight for estimating current and near-term volatility parameters, while long-maturity options matter most for the parameters that govern long-run behavior. This kind of interpretability is one of the practical advantages of using a tree-based method over a neural network.

There are some clear limitations. The dataset has only about 4,000 training samples; scaling up to 50,000 or more would likely sharpen the predictions, especially for κ. Everything in this thesis is tested on synthetic data, which is clean and perfectly consistent with the Heston model. Real market data would introduce noise, model mismatch, and potentially different results. The speed advantage of XGBoost, while theoretically large, is modest in the synthetic setting because Differential Evolution also converges fast on clean data.

Looking ahead, there are several natural extensions. The most important would be validating the approach on actual S&P 500 index options to see how well it transfers from synthetic to real data. Beyond that, one could apply the same framework to richer models like the Bates model (which adds jumps on top of stochastic volatility) or to rough volatility models. Comparing XGBoost head-to-head with neural networks on the same calibration task would also be informative. Finally, adding more market features as inputs, such as the VIX or trading volume, might help improve prediction accuracy.

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Appendices

Appendix A: AI Usage Declaration

I declare that no part of this thesis has been produced by artificial intelligence software.

[Note: The use of software for grammar and spelling checks is consistent with Declaration A.]

YES

I declare that parts of this thesis include contributions from artificial intelligence software and that this is in line with acceptable use as defined in the AI usage guidelines, and consistent with proper academic practice. The content can nevertheless be considered my own words.

I acknowledge the use of AI software for:

(i) Idea generation, structural suggestions, support in understanding core concepts.

[AI tool: Claude]

YES